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delab-trees: A Python library for analysing conversation reply trees in social media research

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Software

- Review
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Summary

`delab-trees` is a Python library for analysing conversational reply trees as they arise in social-media platforms (Twitter, Reddit, Mastodon) and online discussion fora. It treats a *conversation* as a directed reply graph rooted in an opening post, exposes both a tabular (`pandas.DataFrame`) and a graph (`networkx.DiGraph`) representation of that conversation, and provides a uniform API for: validating tree structure, extracting *conversation flows* (root-to-leaf paths through the reply graph), computing author-centrality metrics, comparing flows by quantitative content features, and (experimentally) training simple neural models for participation-aware author scoring. The library is designed to operate on a single shared input schema (`tree_id`, `post_id`, `parent_id`, `author_id`, `text`, `created_at`) so that data drawn from different platforms can be analysed with identical code paths.

Statement of need

Research on online conversations in computational social science is dominated by post-centric or user-centric measures: counts of replies, follower-network influence, sentiment of individual posts. These quantitative reductions overlook the *structural* dimension of online conversations — the shape of the reply tree and the position of authors within it — which carries information about deliberative quality that text-level analysis cannot recover (Aragón et al., 2017; Magnani et al., 2012). What has been missing is a reusable Python library that takes the *conversation flow* (root-to-leaf path) as a first-class object and supports research questions about deliberation, moderation, and participation at that granularity. `delab-trees` fills this gap.

State of the field

Several lines of work argue that the conversation *structure* is itself a meaningful unit of analysis: reply trees and the implicit thread structures that recover them (Wang et al., 2008), branching-process models of Twitter conversations (Nishi et al., 2016), graph reconstruction of Twitter conversation graphs (Cogan et al., 2012), and meso-/macro-level conversation structures in support forums (Joglekar et al., 2020). These contributions establish that reply-tree shape carries analysable signal, but they are realised as study-specific code rather than as a reusable tool. No widely-available Python library takes the conversation flow as a first-class object across platforms; `delab-trees` is, to our knowledge, the first to do so under a single shared schema.

Software design

Two core abstractions structure the library: `TreeManager` holds a dictionary of trees keyed by `tree_id`, and `DelabTree` represents a single conversation, carrying its `pandas.DataFrame` form alongside a networkx reply graph. A DelabTree can be projected into derived graphs (author graph, author-interaction graph, recursive tree), cleaned (cycle removal, orphan attachment, merging consecutive same-author posts), filtered (largest connected component, flows of a given length), and scored (branching weight, depth, root dominance, betweenness/closeness/Katz centrality per author). Building on these abstractions, the library provides:

- A platform-agnostic representation of reply trees from a minimal schema, so that Reddit, Twitter, Mastodon and forum data can be analysed under one API.
- Conversation-flow extraction with optional filtering by length and predicate.
- FlowDuo, a routine for finding the two flows within a conversation with the largest contrast on a chosen content metric (e.g. sentiment) — useful for selecting illustrative cases or constructing matched comparisons.
- Experimental, participation-aware author-scoring models that aim to go beyond raw centrality: a *response-based* (RB) classifier that learns whether an author has likely seen a post from reply-distance features, and a *prediction-based* (PB) model that scores authors by their predictability as the next contributor given conversation context (Dehne, 2023). These two models are provided as a research prototype and are still under active development; the stable, tested core of the library is the reply-tree representation, conversation-flow extraction, FlowDuo, and the centrality metrics described above.

Research impact statement

delab-trees is the shared substrate for several recent studies of deliberation in online discussion. It supports the conversation-flow unit of analysis in a large-scale empirical study of user moderation on Twitter, where reply-tree reconstruction and flow extraction are upstream of every downstream model (Dehne & Gold, 2024). It implements the participation metrics evaluated in the CCCP study comparing Reddit and Twitter (Dehne, 2023). And it underpins the intervention-recovery procedure in a deployed-bot study on Reddit and Mastodon, where deleted bot posts are matched back into recovered reply trees in order to preserve the empirical record for analysis (Dehne, 2024). By providing this shared substrate, the library lowers the cost of structural conversation analysis and makes the published results reproducible: it is distributed via PyPI (pip install delab_trees) and ships with anonymised Reddit and Twitter example datasets for reproduction of the published results.

AI usage disclosure

The functional code of delab-trees was written by hand and quality-checked in the course of the scientific work it supports; the substantive implementation dates to 2023–2024, as the commit history shows, roughly two years before this submission. In preparing the library for JOSS publication, the AI coding assistant Claude Code was used in a limited, non-functional capacity only: cleaning up the documentation and the package metadata files, and assisting with refactoring the dependency declarations and the test pipeline. No part of the library's analytical or algorithmic functionality was authored by an AI system; all such code is hand-written and was validated during the underlying research.

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